AI Foundations | Practical Evidence

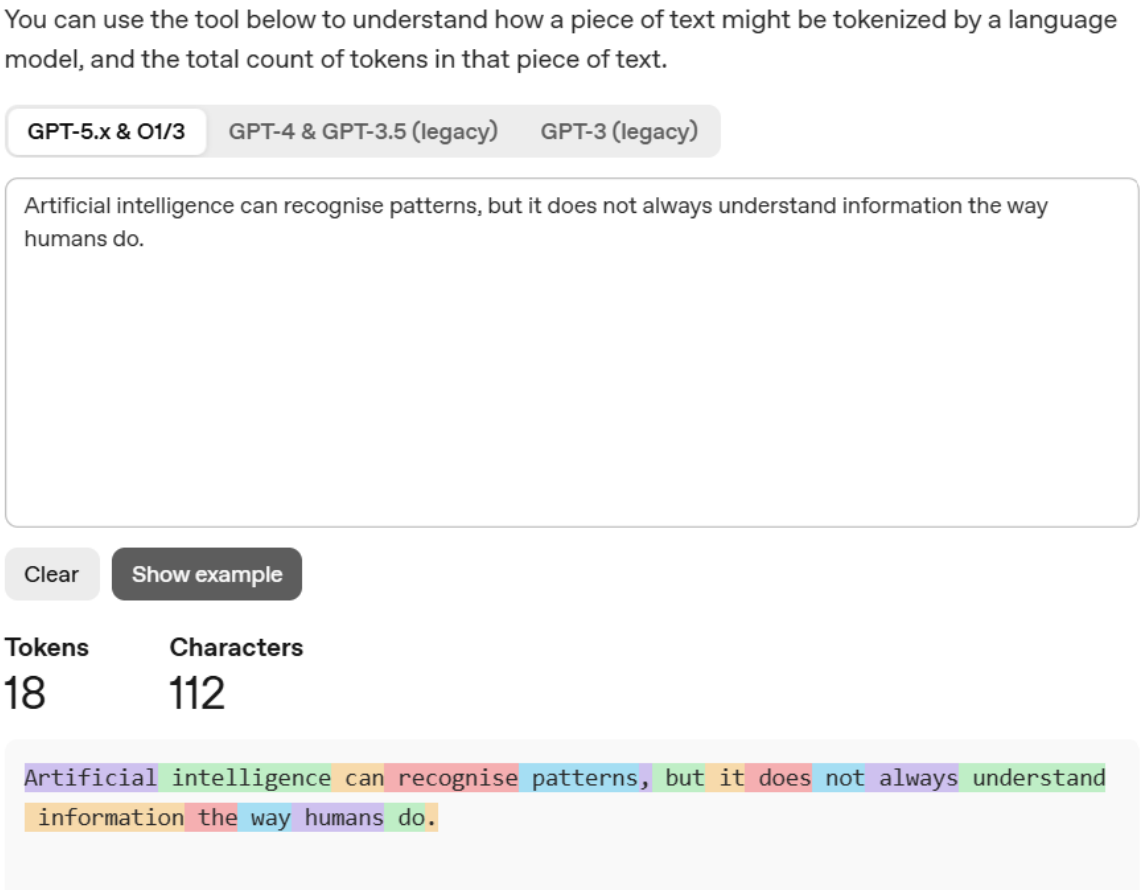
BDAIB Badge Exercise • Four real examples

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01 Token count

Paste a sentence into the OpenAI tokenizer and record the result.

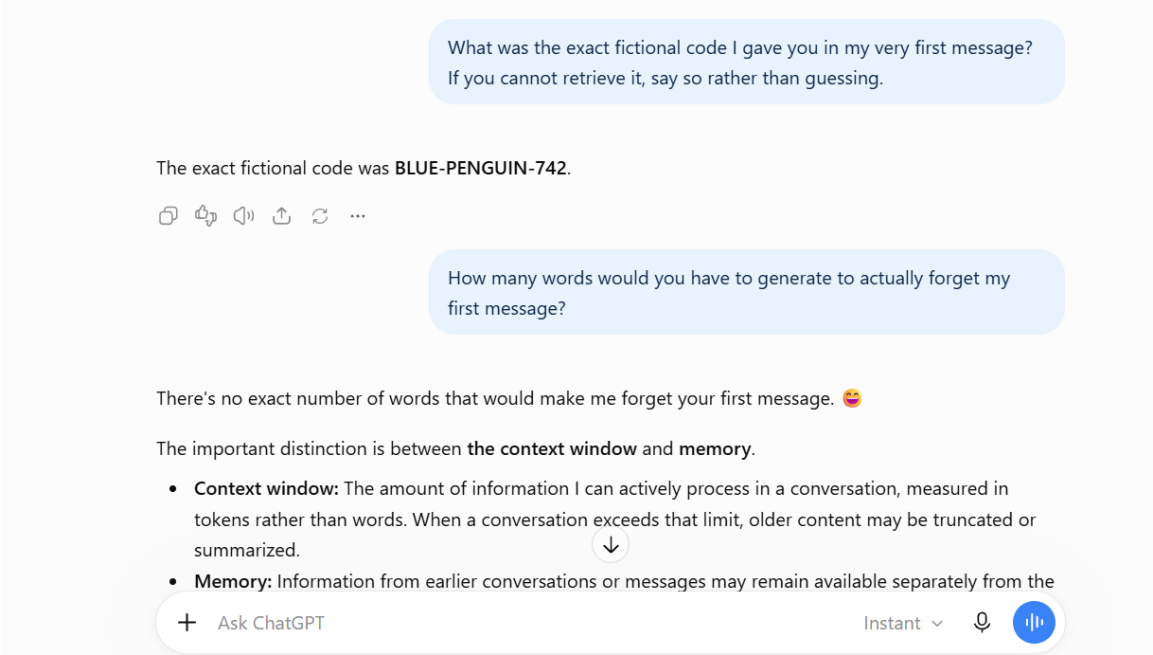
Sentence tested	Artificial intelligence can recognise patterns, but it does not always understand information the way humans do.
Tokenizer / model	GPT-5.x & 01/3
Token count	18
One-sentence explanation	I tested the sentence using Tokeniser from Chat GPT, which split it into 18 tokens.



02 Context overflow

Use a long chat or document where the model could not recover an earlier detail.

Prompt/Method	**I asked ChatGPT to remember a fictional code, BLUE-PENGUIN-742, without repeating it. I then submitted 10 unrelated prompts requesting 3,000-word essays on different topics to test whether the original information would be lost as the conversation grew.**
AI-Response	After the first five prompts, ChatGPT correctly responded: "The exact fictional code was BLUE-PENGUIN-742." After another five lengthy prompts, ChatGPT-6 still recalled the original code correctly.
One-sentence explanation	I attempted to trigger context overflow by extending a conversation with 10 lengthy essay requests, but ChatGPT-6 successfully recalled information from the beginning, meaning no context overflow was demonstrated.

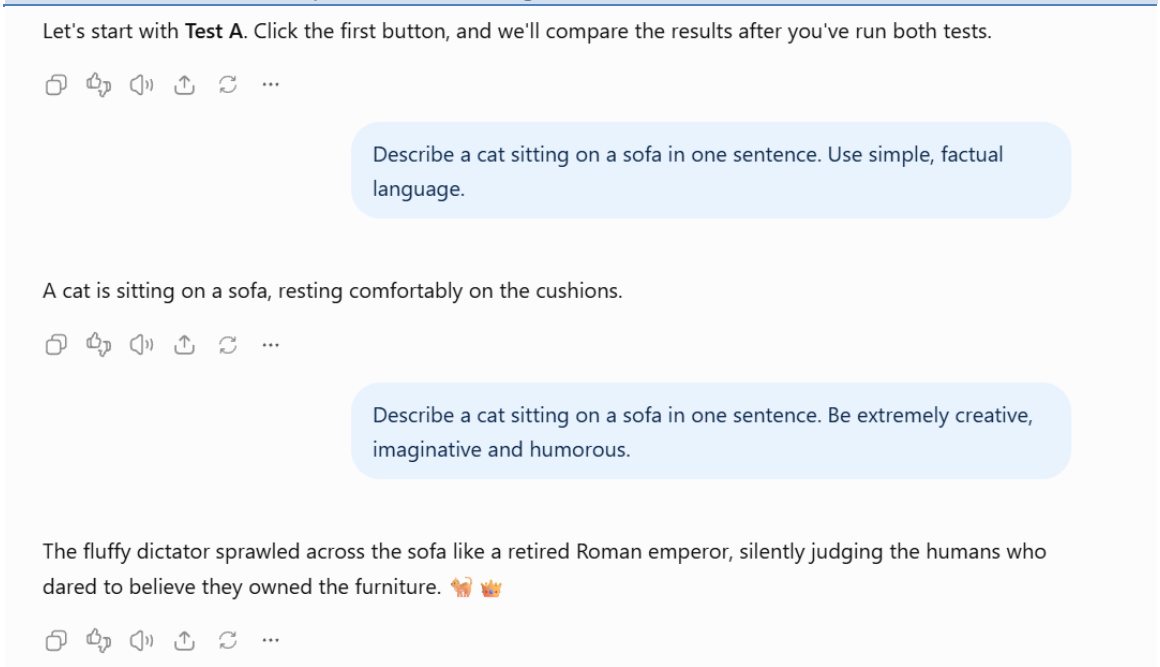


03 Temperature / creativity comparison

Compare two outputs for the same task, using different creativity instructions.

Base task	**Describe a cat sitting on a sofa in one sentence.**
Prompt A / response	P: Describe a cat sitting on a sofa in one sentence. Use simple, factual language. R: A cat is sitting on a sofa, resting comfortably on the cushions.
Prompt B / response	P: Describe a cat sitting on a sofa in one sentence. Be extremely creative, imaginative and humorous. R: The fluffy dictator sprawled

	across the sofa like a retired Roman emperor, silently judging the humans who dared to believe they owned the furniture. 🐈‍⬛👑
One-sentence explanation	I compared two AI responses to the same task using different creativity instructions and observed how the output changed from factual and predictable to imaginative and humorous



04 Hallucination

Find a confident, plausible falsehood and verify it using an independent source.

Example:	**Incorrect Brazilian coat of arms in Flávio Bolsonaro's public letter (October 2026).**
Observation:	An early version of the letter reportedly used a historical Brazilian coat of arms displaying "Estados Unidos do Brazil" instead of the current official emblem. An associate attributed the mistake to ChatGPT, and the document was subsequently corrected.
Hallucination signature:	Confident, plausible falsehood — the document presented an outdated national symbol as if it were the current official one.
One-sentence explanation	"An AI-assisted political letter reportedly used an outdated Brazilian coat of arms as an official symbol, illustrating how plausible-looking AI-generated content can contain factual errors."



INCORRECT VERSION

Historical coat of arms used in the first version of Flávio Bolsonaro's letter (October 2026).

Inscription: "Estados Unidos do Brazil"



CURRENT OFFICIAL VERSION

Official coat of arms currently used by Brazil (Republic established in 1889).

Inscription: "República Federativa do Brasil"

As Brazilian media and political observers speculated about the abrupt change, one of Bolsonaro's closest allies suggested that the explanation for the mistake was not political, but technological: the use of OpenAI's ChatGPT.

Right-wing influencer and podcast host Paulo Figueiredo, who serves as an unofficial spokesperson for the Bolsonaro family in the United States and first shared the letter with journalists, said the document had gone through several drafts and incorporated contributions from multiple people.

Figueiredo said that Bolsonaro's younger brother, former Congressman Eduardo Bolsonaro, sought ChatGPT's help to finalize the letter. According to Figueiredo, the chatbot inserted an outdated version of Brazil's coat of arms and used a previous draft of the letter.

"When generating the document, AI ended up including an earlier version of the text, in which Flavio presented himself as the true 'president-elect'," Figueiredo wrote on X, adding the mistakes were noticed only after he distributed the letter to the press.

According to him, Bolsonaro ultimately opted for the term "leader" for legal and moral reasons. He added that those involved found the mistake "hilarious" and described tweaking the letter as "nothing major."

OpenAI and Bolsonaro's press officers did not immediately reply to requests for comment.

Source: https://www.reuters.com/business/media-telecom/bolsonaro-ally-blames-chatgpt-now-deleted-president-elect-brazil-claim-2026-10-29/?utm_source=chatgpt.com